

The Weibull Distribution

2026-04-17

Introduction

The Weibull distribution generalises the exponential to allow monotone non-constant hazard rates. It is the default parametric model in reliability engineering and a common choice for parametric survival analysis. The shape parameter controls whether hazard is increasing (ageing), decreasing (early failures dominate), or constant (exponential).

Prerequisites

Exponential distribution, hazard and survival functions.

Theory

A Weibull random variable with shape $k > 0$ and scale $\lambda > 0$ has PDF

$$f(x) = \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} \exp\left[-\left(\frac{x}{\lambda}\right)^k\right], \quad x \geq 0.$$

CDF: $F(x) = 1 - \exp\left[-(x/\lambda)^k\right]$.

Hazard function: $h(x) = (k/\lambda)(x/\lambda)^{k-1}$. Three regimes:

- $k < 1$: decreasing hazard (early failures / “infant mortality”).
- $k = 1$: constant hazard (exponential).
- $k > 1$: increasing hazard (wear-out / ageing).

Moments:

- $E[X] = \lambda \Gamma(1 + 1/k)$.
- $\text{Var}(X) = \lambda^2 [\Gamma(1 + 2/k) - \Gamma(1 + 1/k)^2]$.

Accelerated failure time form: $\log T = \mu + \sigma W$ where W is standard extreme-value; the AFT parameters relate to Weibull's $k = 1/\sigma$ and $\lambda = e^\mu$.

Assumptions

Non-negative continuous data, monotone hazard (log-hazard linear in log-time).

R Implementation

```
library(survival)

# PDFs for three shape regimes
x <- seq(0, 5, length.out = 400)
plot(x, dweibull(x, shape = 0.5, scale = 1), type = "l", col = "#F4A261",
      lwd = 2, main = "Weibull PDFs", ylab = "f(x)")
lines(x, dweibull(x, shape = 1, scale = 1), col = "#6A4C93", lwd = 2)
```

```

lines(x, dweibull(x, shape = 2.5, scale = 1), col = "#2A9D8F", lwd = 2)
legend("topright", c("k = 0.5 (decreasing h)", "k = 1 (exponential)", "k = 2.5 (increasing h)"),
      col = c("#F4A261", "#6A4C93", "#2A9D8F"), lwd = 2)

# Simulate and fit
set.seed(2026)
X <- rweibull(500, shape = 2.0, scale = 1.5)
fit <- survreg(Surv(X, rep(1, length(X))) ~ 1, dist = "weibull")
summary(fit)

# Convert survreg output to Weibull parameters
c(shape = 1 / fit$scale,
  scale = exp(coef(fit)))

```

Output & Results

Call: survreg(...)

	Value	Std. Error	z	p
(Intercept)	0.40578	0.01574	25.78	<2e-16
Log(scale)	-0.69312	0.03293	-21.0	<2e-16

Scale = 0.500

shape	scale
2.000	1.501

The fit recovers $k \approx 2$ and $\lambda \approx 1.5$ – the true parameters. The increasing-hazard regime is visible in the PDF.

Interpretation

In survival analysis, a Weibull model is often the starting parametric fit: “we fitted a Weibull regression with shape $k = 1.3$ (95 % CI 1.1-1.5), indicating increasing hazard over follow-up.” Weibull is the default choice when the proportional-hazards assumption holds on a transformed scale.

Practical Tips

- R’s `rweibull(n, shape, scale)` conventions differ from `survreg`, which parameterises via intercept and $\text{scale} = 1/k$.
- For right-censored data, use `survreg(..., dist = "weibull")`, not `fitdistr` on complete data.
- Check the fit via a log-log plot: $\log(-\log S(t))$ vs $\log t$ should be approximately linear.
- If shape estimates are very close to 1 with narrow CI, consider the simpler exponential.
- `flexsurv::flexsurvreg(..., dist = "weibull")` is a modern alternative with richer diagnostics.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots